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Why do educational innovations come and go? What do we know? What can we do?

Robert J. Stevens*

Department of Educational Psychology, The Pennsylvania State University, 202 CEDER Building, University Park, PA 16802, USA

Abstract

This article discusses the nature of innovations in the educational community and how educational changes are made. Recent research documents ongoing change with new innovations coming and going with little or no concern for their efficacy or importance. The article describes some of the possible reasons why education goes through fairly continual change. The article also attempts to offer suggestions on how teachers, teacher trainers, administrators, and educational researchers can have a positive impact on reducing or slowing the processes of innovations coming and going in American schools.

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Over the past 50 years, American education has seen a number of educational innovations sprout in the educational literature, grow in school districts across the country, yet many wither and die a few years later. Many of these innovations have spread or come from other countries, but few countries have experienced the seeming continual change that we in the United States experience. A list of innovations in American education in the past 20 years would be too long for this manuscript, but might include such things as open classrooms, discovery learning, mastery learning, and cooperative learning as more generic innovations. It might also include specific instructional models like the Essential School Model (Sizer, 1987, 1996), Reading Recovery (Pinnell, Lyons, DeFord, Bryk, & Seltzer, 1994), and Success For All (Slavin, 2002; Slavin, Madden, Karweit, Livermon, & Dolan, 1990). Yet, external evalua-

tions and reviews of the research literature about these innovations seem to indicate that their longevity—whether they survive or not—has less to do with their effectiveness than one might think (Datnow, Hubbard, & Mehan, 2002; Gage, 1985; Olson, 1999; Slavin, 2001).

In recent years there has been some discussion of why educational innovations come and go in both national conferences in the United States (e.g., American Educational Research Association) in journals (Alexander, Murphy, & Woods, 1996; Slavin, 1989), and in monographs (Datnow et al., 2002). While there is little disagreement about the existence of the phenomenon, authors have taken different approaches in explaining and understanding it.

Alexander et al. (1996) look more at teachers in an attempt to understand the efficacy of innovations in schools. They propose that teachers tend to implement in their classrooms what they know and understand, in spite of whatever innovation may be adopted by the school, or what evidence

*Tel.: +1-814-863-2417; fax: +1-814-863-1002.

E-mail address: rjs15@psu.edu (R.J. Stevens).

may be offered about their current methods or innovative methods. Alexander et al. (1996) suggest changes in teacher preparation that develop teachers' understanding of learning philosophies, theories and principles. That the preparation instruct teachers in how to apply those principles to increasing student learning, and teaches teachers "more about less" by focusing less on a survey of what exists and more on developing deeper understanding of what is taught. Further, they believe that a deeper understanding of these theories will better prepare teachers to evaluate and understand innovations that they will confront in the future.

Slavin (1989) writes more about the nature of change within schools and the way educational decision-making leads to persistent change for the sake of change. He depicts educational change as a "fashion trend" as opposed to a more analytic or scientific building process. In this process Slavin states that innovations are proposed and supported by a few "gee whiz" testimonials and that innovative school districts latch on to the program and begin the new wave, or next swing of the pendulum. He suggests that the programs spread wildly from there, being picked up by more districts and pushed by administrators and staff developers who do not want to be left behind. As the program spreads, calls for evaluation increase as do the questions or complaints about the program. Slavin argues that by the time we have any substantial, well-designed evaluations of the innovation and data on its effectiveness, the innovative districts have already moved on to a new innovation and have started the pendulum swinging quickly in another direction. Slavin (1989) suggests more emphasis in the field of education on using evaluative metrics on which to base decision-making. This proposal is much like that recommended in President Bush's No Child Left Behind Act of 2001 that emphasizes the need for scientifically based research.

Datnow et al. (2002) take a more complex look at school reform in low achieving schools in the context of reform efforts that are frequently forced by external influences and usually related to high-stakes tests. This type of externally imposed reform certainly shares characteristics described

by Slavin and Alexander and her colleagues, it also is significantly different in terms of change being imposed externally and highly charged with politics and power struggles. They see a dynamic process of implementing change in a complex socio-cultural context within the school and community.

Certainly there is very persuasive reasoning in all of these authors' work, however, I would like to add some areas of discussion to the issue of why innovations come and go, with a focus less on externally mandated change in low achieving school, and more on the cycle of change common to all school in the United States. The change to which I refer is better thought of as "innovation" than externally imposed "reform" and is more in line with what Gage (1985) was discussing in his scientific basis for the art of teaching. This more fundamental notion of change—or lack of it—common to teaching processes in general has broader implications. The stimulus for this discussion can perhaps be summarized by Gage's comment "classroom teaching has turned out to be extremely resistant to the attempts of the last 100 years... to introduce radical changes" (Gage, 1985, p. 3). As Cuban (1984) has suggested, teaching practices have changed little and teachers tend to teach as they were taught. Specifically I propose that teacher-trainers, teachers, administrators, educational researchers, and developers all play a role in the swinging of the educational innovation pendulum, and as such we need to change the domain of each to have a lasting impact on the problem.

1. Teacher training

Much has been written about the need to better prepare teachers for the demands they will meet in classrooms (Griffin, 1999; Pushkin, 2001; Yinger, 1999). With regard to the issue of innovations, there seem to be two areas where teacher preparation could better meet teachers' long-term needs: (a) training with more of a focus on measurable effects in their instruction, and (b) more knowledge about the theory and principles that describe

how learning occurs and how instruction influences learning.

One could easily point to current legislative movements at the national (No Child Left Behind) or state levels (state-wide student assessments) to explain the importance of teacher training focusing on “measurable effects” as an outcome for instruction. It is clear that schools and teachers are going to be held accountable for student performance regardless of what philosophical approach they may take to education. In a more pragmatic sense teachers should *expect* that if they put time or effort into implementing an innovative instructional approach they or their students *benefit* in some measurable way from that effort. As [Griffin \(1999\)](#) suggests, it is important instruction be for the benefit of students, so certainly any change in instruction should also be for students’ benefit.

Given this lack of measurable outcome orientation, teachers are left to other criteria when determining what to do, or what programs to try. The decision-making void may be filled with anything from reason to experience to whimsy. I also find this lack of focus on measurable outcomes in teacher preparation ironic because in the public domain in which teachers work administrators, parents, and legislators place clear and strong demands on teachers for measurable effects, yet teachers often seem inadequately prepared to respond to these demands. Instead teacher training often puts more emphasis on philosophy than on what works ([Deegan, 1995](#)).

Teachers are given only very rudimentary knowledge in areas like learning, instruction, and motivation often covered briefly during a few weeks in an introductory educational psychology class. While learning, instruction, motivation, and assessment are central to what teachers do, their training often provides superficial knowledge about the processes involved in these areas and how teachers can have an impact on them. As [Yinger \(1999\)](#) has suggested, we need to increase the use of the research base in teacher education, and train teachers to know and to use effective teaching practices. This paucity of basic theory and research about how and why students learn, particularly given the wealth of knowledge we have, is appalling. Asking teachers to make

informed instructional decisions based upon this limited knowledge base is like going to a doctor for diagnosis and treatment when he has had only a freshman biology class. And yet teachers make instructional diagnoses and remediation many times a day.

Like the suggestions of [Alexander et al. \(1996\)](#) and [Gage \(1985\)](#) teachers need a more in-depth understanding of the principles of learning and learning theory so they can understand how instruction may have an impact on students’ learning. A similar in-depth understanding of motivation also is necessary. Learning and motivation theory are the underpinning for all the methods taught in teacher preparation programs, but students spend significantly less time learning the theory (why the methods work) than they do the methods themselves. As such it is no surprise when teachers take more of a “bag of tricks” approach to teaching than a rational, theoretically driven approach to making instructional decisions.

2. Teachers

Teachers also play a critical role in swinging of the innovation pendulum in part as a result of the training described above and in part in how they act in the structure of the schools. When it comes to trying new programs there are three kinds of teachers: the innovators, those who like trying new instructional approaches; the immovable, those who refuse to change; and the large middle group that essentially takes a “wait-and-see” attitude. Innovators are often eager to try new programs and seem to thrive on the risks involved. However, it is exactly those risks that seem to inhibit all of the other teachers from changing from what they know and feel comfortable with to the risks of the unknown. It is not unusual for only a minority of teachers to fully adopt and implement new programs. Innovations come and go in part because they never become widely or deeply implemented in a school ([Cuban & Usdan, 2003](#); [Datnow et al., 2002](#)).

Another obstacle to adopting innovations is that teachers are frequently given very little support and reward for changing what they do in class-

rooms (Datnow et al., 2002). When changes are instituted teachers may be left on their own to figure out how to do the innovation, how to develop appropriate curriculum materials, how to mesh curriculum and processes to district or state goals, and how to solve problems specific to the context in which they are implementing. Yet when they accomplish successful implementation there is little recognition or reward for doing so. On the other hand, teachers are likely to risk rebuke when innovations fail or struggle. In this context, is it any surprise that most teachers prefer to “stand pat” with what is comfortable rather than to attempt an innovation, no matter how convincing? An innovation cannot become institutionalized when only a minority of the teachers embrace the reform and fully implement the innovation. It remains “experimental” or novel and without being widely accepted and used, the innovation is bound for eventual rejection (Datnow et al., 2002). Enertia favors a lack of large-scale change.

3. Administrators

Over 20 years ago Ron Edmonds (1979) described effective schools as places where the principal was the instructional leader. However, while administrators (i.e., principals and superintendents) have advanced degrees in education, many have little or no advanced training in learning, instruction, and motivation and often only cursory knowledge about evaluating educational programs (McCarthy, 1999). It would seem that if administrators are to be instructional leaders, they should be highly trained in these areas that are central to what affects and is affected by classroom instruction. Indeed, the National Policy Board for Educational Administrators (NPBEA) recommended as early as 1989 that administrators be trained with more emphasis on teaching and learning processes, school improvement, and methodologies for program evaluation. However, curricula for graduate programs in educational administration continue to be dominated by school law, finance, and organizational development (Duke, 1992; McCarthy, 1999).

While administrators are in the frontline of the debate with communities and government agencies about whether educational programs and schools have any measurable effects on students, they may be woefully unprepared to engage in that discussion. With little or no training in educational evaluation, administrators are seriously handicapped in their attempts to measure the effectiveness of what their schools are doing. A lack of sophistication in research and evaluation also handicaps administrators in making decisions about whether to take on attractive innovations or even which textbook to adopt. The lack of analytic decision-making opens the door for decisions made on a variety of less substantial bases.

Similarly, administrators who lack a thorough understanding of the research as well as a substantial background in learning, motivation, and instructional theory seem to grasp on to specific innovations or ideas rather than understanding the processes or research behind them. New administrators may come into a district with two or three ideas in their bag of tricks that they expect all of the teachers to implement. This implies that the teachers must stop implementing the programs of the previous administrator, again without regard to its efficacy (Slavin, 1989).

A new administrator bringing in a new program might not be so bad except that administrators, particularly superintendents, tend to have a fairly short tenure, often five years or less. In effect they come into a district with a few ideas like intensive scheduling, cooperative learning, learning styles, or outcome-based education and push for widespread implementation. After they have accomplished their changes they leave the district and are replaced by another administrator who brings in his/her own list of “things to do.” Over time teachers have seen administrators come and go. They also have seen new ideas come and go. The teachers *know* that they will be in the school ten or fifteen years from now and that the administrator is likely to be gone. All they have to do is wait, for this too shall pass.

At the same time, there is little reward or excitement for administrators to work on improving or maintaining existing programs.

Maintenance requires a focus on details and small incremental change, two things that are seldom if ever rewarded in education. Rewards tend to go to what is “new” or “innovative” further reinforcing the pendulum’s swing.

4. The school as an institution

It is also important to address the school district as an institution in order to understand the nature of making long-lasting change through educational innovations (Datnow et al., 2002). Most innovations come from outside the school, from developers or researchers. Typically the school contacts the developer or a trainer to come to the school district to teach them how to implement the innovation. However, it is the *developer’s* program, not the school’s program. For a program to have a long-lasting impact on a school the district needs to adopt the program and develop some ownership and investment in the program. The school needs to develop their own resources as the primary source for answering questions and solving problems related to the program. The school or district needs to develop the ability to monitor and improve implementation and maintain the program after the developer or trainer has left. At a minimum that includes knowing: Who will train new teachers who come to the district? Who will create new materials as the textbooks change? Who will coach the teachers on an almost daily basis so they can become proficient and successful? The degree to which schools can rely on someone from the outside to fulfill these needs is very limited. They need to invest in the program by developing their own resources for the maintenance and expansion of the innovation.

For example, in the mid-1970s the Anne Arundel County School District in Maryland decided that they were going to adopt cooperative learning as an educational practice. They institutionalized the process in part by having their staff development personnel become experts and certified trainers in cooperative learning. The district then used their own resources to gradually train and coach all of the teachers in the district and any new teachers who entered the district. This focused

staff development lasted until the mid-1990s, when a change in administration redirected the district’s staff development activities (Wayne Bark, principal at Crofton Elementary School in Anne Arundel County School District, personal communication, July 23, 2003). As a result, for about 20 years the staff development personnel trained and coached their teachers in cooperative learning. While this may seem short, 20 years is truly long term implementation when it comes to educational innovations in the United States.

The commitment of a district to developing their own resources also has an impact on teachers and their willingness and motivation to try an innovation. Quite simply, teachers perceive the district as being serious about an innovation if they commit resources to it. It suggests that the innovation is more permanent and that it may not be just another thing that will come and go in a couple years. As a result teachers may be more inclined to put their time and effort into trying the innovation. Similarly teachers may see the local experts as safe and reliable support they can readily use if they attempt the innovation. As a result they may be more willing to take the risk of trying an innovation.

5. Researchers and developers

Researchers and developers also play a role in the swinging of the pendulum of educational innovation. First research needs to be made more accessible and practical to teachers and administrators. Too often we write and do research for ourselves, our cohort of colleagues in the field. Once research has been published in a prestigious academic journal we move on to the next study, rather attempting to relate the findings to the lives of teachers. Publication in “practitioner” journals is devalued as less scholarly, and often disparaged in promotion and tenure decisions. As a result, important research supported knowledge is not translated for consumption by teachers and administrators. If, as I propose, teaching needs to be built on a foundation of research supported knowledge, then as researchers we need to write

for more audiences, particularly teachers and administrators.

We also need to rethink our approach to research. Perhaps we can benefit from following the model of the sciences and medicine. One of the ways science and medicine accumulate knowledge is through replication of research. Replication studies help substantiate and buttress the importance of knowledge gained through research. While we use scientific methodologies in education we devalue the importance of replications to the point where authors have difficulty in publishing them. We act as though getting a finding once is sufficient to prove its truth. This results in a perpetual push for each study to be something new and in turn we ignore the old, pushing the pendulum a bit more.

When researchers and developers attempt to apply their findings to educational practice, or when we try to implement innovations in schools, we need to realize that implementation requires more than a couple of workshops with teachers. Good implementation of most innovations requires ongoing training, observation, and coaching. Many innovations fail because they are not implemented systematically and maintained through observation and coaching over time. (Administrators also share the guilt for bringing people in for one-day workshops without any consideration of follow-up. These kinds of “dog and pony shows” seldom promote substantial change.) In many ways it would be better if the school and teachers did not take on the innovation at all. A one-day workshop in an innovation sets teachers up to fail. The failure is then attributed to the ineffectiveness of the innovation, rather than to the ineffectiveness of the training and implementation effort.

For example, the Success for All program has taken steps to resolve this problem (Cooper, Slavin, & Madden, 1998; Slavin & Madden, 1999). When schools commit to implementing Success for All, they also agree to fund both short-term and long-term follow-up by the project staff. During the first year Success for All sites are frequently monitored and coached by the project staff. Even after the first year there are twice-a-year observations by Success for All staff to help

maintain the integrity of the program. After each of the observations teachers are given formative feedback designed to help them to improve their implementation of the various components of the model. In an attempt to institutionalize the program in the school, the Success for All developers have committed themselves and the district to systematic and intensive staff development.

Program developers also need to help schools institutionalize the program, as described above. They need to develop school district personnel as local experts in the program with detailed knowledge about the program’s instructional processes. The local expert, or facilitator, becomes indispensable in maintaining the program as they actually have the time to focus on the details of implementation, something administrators often do not have the time to do. Also these facilitators typically are faculty members and their observations are seen as less threatening and evaluative than those by administrators.

6. Technical assistance centers

Finally, I believe we need to create institutions to facilitate the dissemination of empirically based research to the educational community. It is unrealistic to expect researchers—who are frequently full-time university faculty members—to have the time and resources to disseminate their research. This is particularly important now that the federal government is concerned with “ramping up” effective applied research. We need institutions that can translate research into practice and disseminate effective applied research to schools across the country. Many states currently have technical assistance centers that serve this purpose, particularly in special education. They act as an interface between the schools and the researchers, providing the valuable service of transmitting knowledge broadly in the educational community.

A technical assistance center has the advantage of having close contact with local schools in their service area, and can establish an effective means of communication with each school district. At

best, most researchers have contact with only a few districts that are somewhat close geographically. Technical assistance centers can also develop expertise in staff development for teachers, which frequently is not the purview of educational researchers. While researchers may have good skills for research, they may not have the skills to offer good training workshops to teachers. Finally, most innovations require ongoing staff development to maintain the model (Cooper et al., 1998). Local experts, like those who could be available through a technical assistance center, could be more responsive to the needs of local districts and would be better situated to coach the teachers and district through the development of their implementation. Perhaps one of the most effective way for the federal government to promote school improvement through empirically based research would be through funding at a national level technical assistance centers to help provide dissemination of innovations that work.

7. Conclusions

Why do innovations come and go in American education? Clearly there are lots of reasons, more than I have described.

What can we do about it? In my view there are many things that can and need to be done to alter this process of change. No single change will suffice.

All those who participate in educating our students—teachers, teacher trainers, administrators, researchers, and policy makers—need to be better informed about the accumulation of knowledge in education. While the issues discussed in this article are largely prevalent in the United States, the need to bring more of a scientific approach to educational practice and decision making is relevant for all countries. Decisions at all levels, from the classroom to the district to the governmental, need to be based upon our accumulated knowledge and evidence of impact. We need to continue to move education on the path to a more evidence based discipline, with an emphasis on building and using a knowledge base to move the field forward. We need to educate practitioners

at all levels in the importance of evidence and train them in ways to use this knowledge. And we need to focus on research questions that have relevance and importance in the lives of learners. In this way we may be able to slow the swing of the pendulum of change.

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